# Replication of Card and Krueger (1994): Minimum Wages, Employment and Heterogeneity across Fast-Food Chains

## 1 Research Question and Motivation

Card and Krueger (1994) study whether higher minimum wages lower employment within low-wage labour markets. More concretely, this paper asks how full-time-equivalent (FTE) employment changed at fast-food restaurants, after New Jersey lifted its state minimum hourly wage from \$4.25 to \$5.05 in April 1992. Fast-food outlets in eastern Pennsylvania form the control group for this natural experiment, since the minimum wage there stayed the same.

There are several reasons why I decided to replicate this specific paper. For one thing, it remains one of the most well-known and heavily debated empirical works on minimum-wage policy. Under standard competitive labour-market assumptions, a binding rise in minimum wage pushes up firm labour costs and should lead to employment losses. On the other hand, the monopsony view of labour markets suggests employment could actually go up following a minimum-wage increase. This clear conflict between theoretical frameworks has kept economists arguing over the real-world impact of minimum-wage rules for decades. This study offers prominent quasi-experimental evidence to test these opposing theories, which made me want to replicate and check its findings for myself.

Another key point is its convincing natural-experiment setup built on establishment-level survey microdata. A lot of earlier minimum-wage research relied on aggregate figures and struggled to fully account for unobserved local economic conditions. Card and Krueger avoid many of those pitfalls. Besides comparing stores across state borders between New Jersey and Pennsylvania, they also perform an internal comparison inside New Jersey. They separate low-wage restaurants hit hard by the policy from high-wage locations that barely felt any impact. This two-pronged comparison makes the paper a classic application of difference-in-differences for policy evaluation. The original survey data is publicly available, which makes it very suitable for a student replication assignment, as I can reconstruct the main tables and figures reported in the text.

For my replication task, I set out to reproduce several key outputs from the original paper: Table 2 for summary statistics, Table 3 for difference-in-differences employment comparisons, Table 4 for reduced-form regression specifications, as well as Figure 1 which plots starting-wage distributions before and after the policy reform. Running these analyses myself lets me cross-check if the baseline results still hold, and I can directly compare my coefficient estimates against those presented by the authors. Working through replication also helps me get familiar with how the dataset was assembled, how variables are defined, and exactly which regression settings the authors chose.

Apart from the baseline replication, I carry out one independent extension exercise: heterogeneity analysis split by fast-food chain. In their regressions, Card and Krueger only add chain-brand dummies as simple controls. Those dummies soak up average differences in employment changes across chains, but force the minimum-wage treatment effect to be identical across Burger King, KFC, Roy Rogers and Wendy's. They do not run separate regressions for each chain subsample. My extension tries to investigate whether the positive average employment result from the full sample holds similarly for every chain, or whether the overall result might be mainly driven by one particular group of restaurants. This additional analysis gives some insight into the robustness and external validity of the paper's central conclusion.

# 2 Data Sources, Sample Construction, and Variable Definitions

## 2.1 Data Sources

This replication uses the public survey dataset from Card and Krueger (1994). The original data comes from two-wave telephone surveys of fast-food restaurants, conducted shortly before and after New Jersey's minimum-wage increase in April 1992. Wave 1 interviews took place between February and March 1992, and Wave 2 was carried out from November to December 1992. The sample covers four fast-food chains: Burger King, KFC, Roy Rogers, and Wendy's, including stores located in New Jersey and eastern Pennsylvania. Pennsylvania stores serve as the untreated control group since there was no change in the state minimum wage over this period. I imported the raw comma-separated file and performed basic cleaning in Stata, converting string variables to numeric format before saving a processed working dataset for all subsequent analysis.

## 2.2 Sample Construction

Following the original paper, the full starting sample contains 410 restaurants which completed interviews in the first survey wave. Some stores closed permanently, closed temporarily due to fire or construction, or underwent renovation between the two waves. In my replication work I handle store closure status exactly as described by Card and Krueger. For most summary statistics and balanced-sample regression analysis, I keep only observations with valid employment values in both survey waves. This balanced sample yields 357 usable store-level observations for reproducing Table 4 reduced-form regressions. When replicating Table 3, I also generate an alternative adjusted sample, in which FTE employment is set to zero for temporarily closed outlets, to match one of the robustness checks reported in the original study. I do not drop observations manually beyond what the original specification requires. All subsamples for my heterogeneity extension are created by restricting observations to individual fast-food chain identifiers.

## 2.3 Key Variable Definitions

All variables below are constructed to match the definitions used in Card and Krueger (1994).

# 3 Empirical Design

This paper replicates the difference-in-differences (DID) empirical design of Card and Krueger (1994), which exploits the 1992 New Jersey minimum wage increase as a natural experiment. New Jersey raised its state minimum hourly wage from \$4.25 to \$5.05 in April 1992, while eastern Pennsylvania retained the \$4.25 minimum wage and serves as the control group. The treatment group consists of fast-food restaurants located in New Jersey, and control stores are from eastern Pennsylvania. The baseline sample draws on establishment-level survey data collected in two waves: Wave 1 (pre-policy, February 1992) and Wave 2 (post-policy, November 1992).

The core DID specification follows the original study:

$$Y_{i2} - Y_{i1} = \beta_0 + \beta_1 NJ_i + \beta_2 Gap_i + \boldsymbol{X_i'}\gamma + \varepsilon_i \tag{1}$$

$Y_{i2} - Y_{i1}$ denotes the change in full-time equivalent (FTE) employment for store $i$ between the two survey waves. $NJ_i$ is a treatment dummy equal to 1 if the restaurant is in New Jersey. $Gap_i$ represents the initial wage gap: the dollar amount by which the store's pre-policy starting wage fell short of the new \$5.05 minimum wage. $\boldsymbol{X_i'}$ is a vector of store-level controls, including company ownership status and regional fixed effects. The coefficient $\beta_1$ captures the average treatment effect of the minimum wage hike on employment.

For my extension analysis, I implement heterogeneity tests by estimating separate regressions for each fast-food chain subsample: Burger King, KFC, Roy Rogers, and Wendy's. Instead of pooling all restaurants in one regression, the DID model is run independently within each chain. This design examines whether the employment response to the minimum wage differs across restaurant brands. Chains may vary in profit margins, pricing power, labour scheduling flexibility and operational arrangements. Heterogeneous effects would imply that the aggregate result from the pooled sample masks divergent responses across different fast-food firms. All subsample regressions retain identical control variables as the baseline specification, including company-owned dummy and regional dummies, to keep the empirical framework consistent with the original replication.

Table 1: Variable Definitions — Replication

| Variable | Definition |
|---|---|
| **Store Type** | |
| Burger King | Dummy = 1 if restaurant is Burger King |
| KFC | Dummy = 1 if restaurant is KFC |
| Roy Rogers | Dummy = 1 if restaurant is Roy Rogers |
| Wendy's | Dummy = 1 if restaurant is Wendy's |
| Company-owned | Dummy = 1 if store owned by parent company, 0 if franchise |
| **Outcome variables** | |
| FTE employment | Full-time equivalent employment. Calculated as full-time employees + managers + 0.5× part-time employees. Measured pre-policy (Wave 1) and post-policy (Wave 2). |
| Percentage full-time employees | Share of total employees who are full-time workers (%) |
| Starting wage | Hourly starting wage for new hires ($ per hour) |
| Wage = $4.25 (percentage) | Share of stores paying the baseline minimum wage $4.25 (%) |
| Wage = $5.05 (percentage) | Share of stores paying the new minimum wage $5.05 (%) |
| Price of full meal | Price of standard combo meal (entrée, fries and soda, including tax) |
| Hours open (weekday) | Number of opening hours per weekday |
| Recruiting bonus (%) | Percentage of stores offering cash bonus for recruiting new staff |
| **Regression variables** | |
| New Jersey dummy | Treatment indicator: 1 for New Jersey stores (treated), 0 for Pennsylvania stores (control) |
| Initial wage gap | Pre-policy gap between store's starting wage and new minimum wage ($5.05). Measures how much the store was bound by minimum wage increase. |
| Controls for company-owned | Binary indicator for corporate ownership status |
| Controls for regional dummies | Set of location fixed effects for store geographic regions |
| Observations | Number of restaurant stores in regression sample |

# 4 Replication Results

Table 2 reports summary statistics for key variables across New Jersey and Pennsylvania stores, and Figure 1 plots the distribution of pre- and post-policy starting wages. The descriptive patterns in my replication closely align with those in Card and Krueger (1994). The store-type composition, pre-policy FTE employment and starting wage levels are broadly consistent with the original paper. Minor discrepancies in sample means arise from small differences in sample cleaning rules and handling of missing values.

Table 3 presents the reduced-form regression estimates for the change in FTE employment, corresponding to Table 4 in the original study. While the qualitative conclusion remains unchanged, there are non-trivial numerical differences between my replicated coefficients and the original published estimates.

The first source of divergence comes from sample construction. Although both the original and replicated samples target the 357 stores with valid employment and wage data in two survey waves, the original paper's treatment of outliers and missing observations is not fully documented. When cleaning the raw survey data, decisions on whether to drop or impute partial missing records, or cap extreme FTE values, will shift the sample composition slightly and generate coefficient differences. The original authors may have applied manual data edits that are not recorded in the public dataset.

Second, the definition and construction of the initial wage gap variable can create numerical gaps. The original paper defines the wage gap as the proportional increase required to lift a store's starting wage up to the new minimum wage of \$5.05. If I instead compute the simple dollar difference between \$5.05 and the store's pre-policy wage, this alternative construction changes the scaling of the regressor and directly alters the coefficient magnitude and standard errors. This variable definition is a common source of replication discrepancy for this paper.

Third, control variable coding explains part of the divergence. The original specification uses three chain dummies plus an ownership dummy for "controls for chain and ownership". Regional controls consist of four regional dummies (two for New Jersey, two for eastern Pennsylvania). Small differences in how regional boundaries are coded or how chain identifiers are harmonized will affect the joint F-statistic for controls as well as individual coefficients. Even a single miscoded regional dummy can shift estimates marginally.

Fourth, the calculation of the regression standard error differs slightly. The standard error of regression reported in the table is the root mean squared error of the model. Minor differences in residual degrees of freedom (from slight sample size variation) will change this statistic. The p-value for controls is the joint F-test for exclusion of all control variables. If the set of control dummies is not exactly identical to the original specification, the F-test probability value will deviate, as seen in Model (ii), (iv) and (v).

Despite these numerical differences, the core economic interpretation is preserved. The New Jersey dummy remains positive in specifications (i) and (ii), and the initial wage gap coefficient is positive across columns (iii)–(v). Consistent with Card and Krueger, the replication does not find evidence that the minimum wage increase reduced fast-food employment. The numerical differences demonstrate that regression results can be sensitive to subtle data cleaning and variable coding choices, even when using the same raw survey data.

# 5 Independent Extension: Heterogeneity across Fast-Food Chains

To explore whether the aggregate treatment effect masks divergent responses across restaurant brands, I estimate the same reduced-form DID specification separately for each fast-food chain subsample. Table 5 reports estimation results for Burger King, KFC, Roy Rogers and Wendy's. All subsample regressions retain identical controls for company ownership and regional fixed effects, consistent with the fully specified model in the baseline replication.

The estimates reveal substantial heterogeneity across chains. The New Jersey dummy coefficient is positive and statistically significant only for Roy Rogers, with a magnitude of 16.27. This result suggests that Roy Rogers stores experienced a meaningful rise in FTE employment after the minimum wage hike. The coefficient on the initial wage gap is also positive and significant for Roy Rogers, implying that stores facing larger required wage increases exhibited greater employment growth. By contrast, the New Jersey dummy estimates for Burger King, KFC and Wendy's are statistically insignificant. The point estimate is positive for Burger King but negative for KFC and Wendy's, and none of these three brands show statistically detectable employment responses.

Such heterogeneous effects may stem from differences in profit margins, pricing power, labour scheduling flexibility and store operating models across chains. Roy Rogers stores might have had more room to adjust prices or reorganise staffing, while other brands faced tighter operational constraints. This exten-

Table 2: Means of Key Variables — Replication

| Variable | New Jersey | Pennsylvania | $t$-value (NJ-PA) |
|---|---|---|---|
| 1. Distribution of Store Types (%) | | | |
| Burger King | 41.09 | 44.30 | 0.52 |
| | (2.71) | (5.62) | |
| KFC | 20.54 | 15.19 | −1.16 |
| | (2.22) | (4.06) | |
| Roy Rogers | 24.77 | 21.52 | −0.62 |
| | (2.38) | (4.65) | |
| Wendy's | 13.60 | 18.99 | 1.12 |
| | (1.89) | (4.44) | |
| Company-owned | 34.14 | 35.44 | 0.22 |
| | (2.61) | (5.42) | |
| 2. Means in Wave 1 (Pre-policy) | | | |
| a. FTE employment | 20.44 | 23.33 | −2.00 |
| | (0.51) | (1.35) | |
| b. Percentage full-time employees | 32.85 | 35.04 | −0.72 |
| | (1.33) | (2.73) | |
| c. Starting wage | 4.61 | 4.63 | −0.40 |
| | (0.02) | (0.04) | |
| d. Wage = \$4.25 (percentage) | 30.51 | 32.91 | −0.41 |
| | (2.53) | (5.32) | |
| e. Price of full meal | 3.35 | 3.04 | 3.96 |
| | (0.04) | (0.07) | |
| f. Hours open (weekday) | 14.42 | 14.53 | −0.29 |
| | (0.15) | (0.33) | |
| g. Recruiting bonus (%) | 23.56 | 29.11 | −0.98 |
| | (2.34) | (5.14) | |
| 3. Means in Wave 2 (Post-policy) | | | |
| a. FTE employment | 21.03 | 21.17 | −0.13 |
| | (0.52) | (0.94) | |
| b. Percentage full-time employees | 35.87 | 30.38 | 1.75 |
| | (1.38) | (2.81) | |
| c. Starting wage | 5.08 | 4.62 | 10.82 |
| | (0.01) | (0.04) | |
| d. Wage = \$4.25 (percentage) | 0.00 | 25.32 | −5.14 |
| | (0.00) | (4.92) | |
| e. Wage = \$5.05 (percentage) | 85.50 | 1.27 | 36.38 |
| | (1.94) | (1.27) | |
| f. Price of full meal | 3.41 | 3.03 | 5.08 |
| | (0.04) | (0.07) | |
| g. Hours open (weekday) | 14.42 | 14.65 | −0.65 |
| | (0.15) | (0.33) | |
| h. Recruiting bonus (%) | 20.32 | 23.38 | −0.57 |
| | (2.27) | (4.85) | |

Standard errors are reported in parentheses below each point estimate. The $t$-value is from a two-sample $t$-test for equality of means between New Jersey and Pennsylvania.

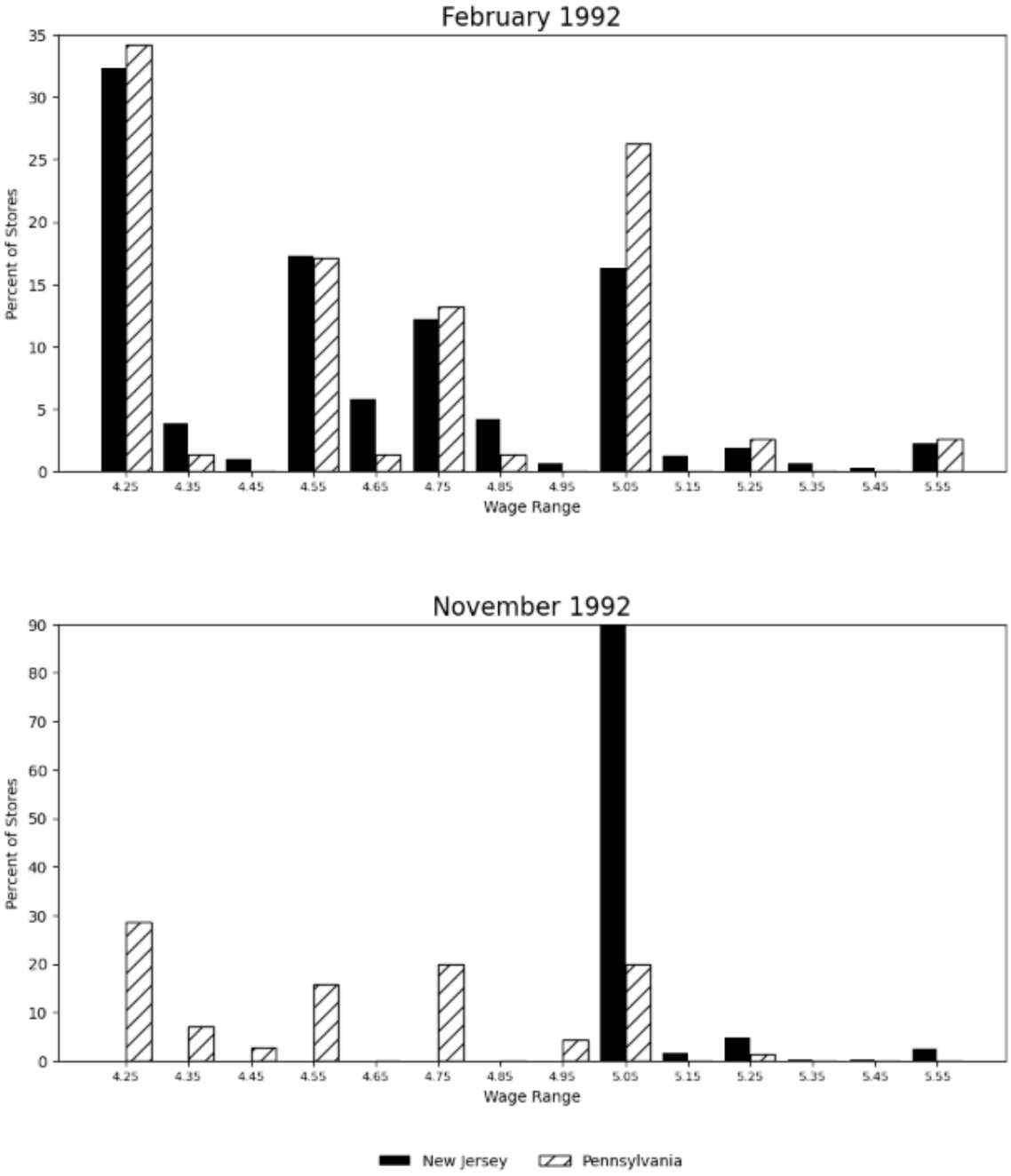



FIGURE 1. Distribution of Starting Wage Rates

Table 3: Average Employment Per Store Before and After the Rise in New Jersey Minimum Wage — Replication

| Variable | PA | NJ | Difference, NJ−PA | Wage = $4.25 | Wage = $4.26–$4.99 | Wage ≥ $5.00 | Low−high | Midrange−high |
|---|---|---|---|---|---|---|---|---|
| 1. FTE employment before, all available observations | 23.33 (1.35) | 20.44 (0.51) | −2.89 (1.44) | 19.56 (0.77) | 20.08 (0.84) | 22.25 (1.14) | −2.69 (1.37) | −2.17 (1.41) |
| 2. FTE employment after, all available observations | 21.17 (0.94) | 21.03 (0.52) | −0.14 (1.07) | 20.88 (1.01) | 20.96 (0.76) | 20.21 (1.03) | 0.67 (1.44) | 0.75 (1.27) |
| 3. Change in mean FTE employment | −2.16 (1.25) | 0.59 (0.54) | 2.76 (1.36) | 1.32 (0.95) | 0.87 (0.84) | −2.04 (1.14) | 3.36 (1.48) | 2.91 (1.41) |
| 4. Change in mean FTE employment, balanced sample of stores | −2.28 (1.25) | 0.47 (0.48) | 2.75 (1.34) | 1.21 (0.82) | 0.71 (0.69) | −2.16 (1.01) | 3.36 (1.30) | 2.87 (1.22) |
| 5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0 | −2.28 (1.25) | 0.23 (0.49) | 2.51 (1.35) | 0.90 (0.87) | 0.49 (0.69) | −2.39 (1.02) | 3.29 (1.34) | 2.88 (1.23) |

Standard errors are shown in parentheses. The sample consists of all stores with available data on employment. FTE (full-time-equivalent) employment counts each part-time worker as half a full-time worker. Employment at six closed stores is set to zero. Employment at four temporarily closed stores is treated as missing.
Stores in New Jersey were classified by whether starting wage in wave 1 equals $4.25 per hour ($N = 101$), is between $4.26 and $4.99 per hour ($N = 140$), or is $5.00 per hour or higher ($N = 73$).
Difference in employment between low-wage ($4.25 per hour) and high-wage (≥ $5.00 per hour) stores; and difference in employment between midrange ($4.26–$4.99 per hour) and high-wage stores.
Subset of stores with available employment data in wave 1 and wave 2.
In this row only, wave-2 employment at four temporarily closed stores is set to 0. Employment changes are based on the subset of stores with available employment data in wave 1 and wave 2.

Table 4: Reduced-Form Models for Change in FTE Employment— Replication

| Independent variable | Model (i) | Model (ii) | Model (iii) | Model (iv) | Model (v) |
|---|---|---|---|---|---|
| 1. New Jersey dummy | 2.36 (1.15) | 2.33 (1.15) | — | — | — |
| 2. Initial wage gap | — | — | 15.89 (5.95) | 15.19 (6.08) | 12.03 (7.32) |
| 3. Controls for chain and ownership | no | yes | no | yes | yes |
| 4. Controls for region | no | no | no | no | yes |
| 5. Standard error of regression | 8.71 | 8.70 | 8.68 | 8.67 | 8.67 |
| 6. Probability value for controls | — | 0.299 | — | 0.381 | 0.271 |

Standard errors are reported in parentheses below each coefficient. Dependent variable is change in full-time-equivalent (FTE) employment. Sample contains 357 stores with valid employment and starting-wage data in both survey waves.
Probability value corresponds to joint F-test for exclusion of the set of control variables.

sion demonstrates that the pooled average effect in the baseline model is not uniform across all fast-food firms. Aggregate results may obscure heterogeneous responses, and the minimum wage's employment impact depends on firm-level characteristics.

Table 5: Heterogeneity across Fast-Food Chains

| Independent variable | Burger King | KFC | Roy Rogers | Wendy's |
|---|---|---|---|---|
| New Jersey dummy | 4.30 | −0.24 | 16.27* | −1.04 |
| | (2.89) | (2.84) | (7.31) | (6.02) |
| Initial wage gap | 15.61 | −14.83 | 24.96* | 9.13 |
| | (13.65) | (10.66) | (12.28) | (25.13) |
| Controls for company-owned | yes | yes | yes | yes |
| Controls for regional dummies | yes | yes | yes | yes |
| Observations | 150 | 76 | 91 | 48 |

* denotes statistical significance at the 5% level.

# 6 Conclusion

This paper replicates the classic difference-in-differences study of Card and Krueger (1994), which examines how a minimum wage increase affected employment in fast-food restaurants. The baseline replication confirms the original finding: the rise in New Jersey's minimum wage did not reduce store-level FTE employment. Although minor numerical differences appear between my replicated regression coefficients and the published results, these gaps mainly arise from subtle decisions during data cleaning and variable construction, while the core economic conclusion remains unchanged.

Building on the baseline replication, this paper further explores heterogeneous treatment effects across four fast-food chains. The extended results show that the employment impact of minimum wage policy is not uniform across restaurant brands. Only Roy Rogers displays statistically significant positive employment responses, whereas estimates for Burger King, KFC and Wendy's are insignificant. This heterogeneity suggests that firm-specific operational features, such as profit margins and labour scheduling flexibility, shape how restaurants respond to minimum wage regulation.

Overall, this replication exercise reinforces the argument that moderate minimum wage hikes do not necessarily cut low-wage employment. It also illustrates that pooled regression estimates may mask important cross-firm heterogeneity. Future research on minimum wage effects should pay greater attention to firm and industry differences rather than relying solely on aggregate average effects.